Linear Fractional Maps in Several Complex Variables

Michael R. Pilla

1. Background

The required background for these problems are courses in complex analysis and linear algebra along with a healthy dose of mathematical maturity.

1.1. Linear Fractional Maps and the Cross Ratio in One Variable. Recall that a linear fractional map (LFM), also known as a Mobius transformation, is defined as

$$f(z) = \frac{az + b}{cz + d} \tag{1.1}$$

where the coefficients a, b, c, and d are complex numbers such that $ad - bc \neq 0$ (otherwise $f(z)$ is a constant function). One important fact about LFMs is that they are 3-transitive. This means that given three distinct points in the extended complex plane (the complex plane plus the point at infinity), there exists a unique linear fractional map that takes these points, respectively, to any other three distinct points in the extended complex plane. This allows one to map domains such as the unit disk to one of different character such as the right half-plane. The standard way to determine a LFM tht adoes this is to use the cross ratio. We define the cross ratio as follows.

Definition 1.1. Given four finite distinct points z_1, z_2, z_3, and z_4 in the complex plane, the cross ratio is defined as

$$(z_1, z_2, z_3, z_4) = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}. \tag{1.2}$$

If $z_i = \infty$, the point at infinty, for some $i = 1, 2, 3, 4$ then we cancel the terms with z_i. For example, if $z_1 = \infty$, then

$$(z_1, z_2, z_3, z_4) = \frac{(z_2 - z_4)}{(z_2 - z_3)}. \tag{1.3}$$

With this definition, we may view the cross ratio as a function of z given by (z, z_2, z_3, z_4).

It is well known that the cross ratio is preserved under linear fractional maps. Thus if ϕ is a linear fractional map in one complex variable and $\phi(z_i) = w_i$ for $i = 1, 2, 3$ then

$$\frac{(w-w_2)(w_1-w_3)}{(w-w_3)(w_1-w_2)} = \frac{(z-z_2)(z_1-z_3)}{(z-z_3)(z_1-z_2)} \tag{1.4}$$

where $w = \phi(z)$. In such a case one may solve for w to determine the map explicitly.

1.2. The Linear Algebra Link in One Variable. We define the associated matrix m_ϕ of the linear fractional map $\phi(z) = \frac{az+b}{cz+d}$ to be given by

$$m_\phi = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Now m_ϕ is a linear transformation acting on *two* variables, while ϕ is a function of one variable So what's going on here? It is well worth taking some time to digest the following interpretation as it is the key to finding LFM results in several variables.

We make the following association. Given a point $u = (u_1, u_2)$ where $u_1, u_2 \in \mathbb{C}$, $u \neq 0$, we declare it to be associated with the point $\frac{u_1}{u_2} \in \mathbb{C}$. For $u_1 \neq 0$ and $u_2 = 0$, our associated point is the point at infinity. Notice that for a scalar c, the point (u_1, u_2) is associated with the same point as $c(u_1, u_2)$. Thus we are associating complex lines in two variables with points in one variables. In fact, in terms of real numbers, projecting straight lines to points is precisely what perspective geometry, the geometry of accurately representing 3D scenes in 2D, is all about. This associated space is known as the complex projective space $\mathbf{CP}^1$ and we call elements such as u the homogenous coordinates. We will write $z \sim u$ if $z \in \mathbb{C}$ is the point associated with $u \in \mathbf{CP}^1$. In $\mathbb{C}$, one may show that linear fractional maps are precisely the linear transformations acting on homogeneous coordinates in $\mathbb{C}^2$.

Furthermore, if $\phi(z) = w$ and the point $u \in \mathbb{C}^2$ is associated with z, then $v = m_\phi u$ is associated with the point w and vice versa. A routine calculation also shows that $m_{\phi_1 \circ \phi_2} = m_{\phi_1} m_{\phi_2}$ and $m_{\phi^{-1}} = (m_\phi)^{-1}$. That is, if $z \sim u$, then $\phi(z) \sim m_\phi u$. Thus, we can toggle back and forth between the complex space $\mathbb{C}$ and $\mathbf{CP}^1$ at our convenience, turning question about LFMs into questions about linear algebra, trading function composition for matrix multiplication.

1.3. The Cross Ratio in Homogeneous Coordinates. We next define the cross ratio in homogeneous coordinates. This holds the key to defining it in several variables. First we define the following quantity. If $u = (u_1, u_2)$ and $v = (v_1, v_2)$ are points in $\mathbf{CP}^1$, then we define

$$[u, v] = \det \begin{pmatrix} u_1 & v_1 \\ u_2 & v_2 \end{pmatrix}.$$

We may then define the cross ratio as follows.

Definition 1.2. Let u_1, u_2, u_3, and u_4 be four distinct points in $\mathbf{CP}^1$, then we define the cross ratio to be

$$(u_1, u_2, u_3, u_4) = \frac{[u_1, u_3][u_2, u_4]}{[u_1, u_4][u_2, u_3]}.$$

We use the same notation as above because as we will observe, this agrees with the above definition of the cross ratio. For finite points we can see this as follows. Given four finite distinct points z_1, z_2, z_3, and z_4 in $\mathbb{C}$, we write the associated points as $u_i = (z_i, 1)$ for $i = 1, 2, 3, 4$ and find

$$(z_1, z_2, z_3, z_4) = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)} = \frac{\det\begin{pmatrix} z_1 & z_3 \\ 1 & 1 \end{pmatrix} \det\begin{pmatrix} z_2 & z_4 \\ 1 & 1 \end{pmatrix}}{\det\begin{pmatrix} z_1 & z_4 \\ 1 & 1 \end{pmatrix} \det\begin{pmatrix} z_2 & z_3 \\ 1 & 1 \end{pmatrix}}$$

$$= \frac{[u_1, u_3][u_2, u_4]}{[u_1, u_4][u_2, u_3]} = (u_1, u_2, u_3, u_4).$$

If one of our points is associated with the point at infinity, then without loss of generality, we let $z_1 = \infty$ and the point associated with z_1 is $u_1 = (1, 0)$ which gives us

$$(z_1, z_2, z_3, z_4) = \frac{(z_2 - z_4)}{(z_2 - z_3)} = \frac{\det\begin{pmatrix} z_2 & z_4 \\ 1 & 1 \end{pmatrix}}{\det\begin{pmatrix} z_2 & z_3 \\ 1 & 1 \end{pmatrix}}$$

$$= \frac{\det\begin{pmatrix} 1 & z_3 \\ 0 & 1 \end{pmatrix} \det\begin{pmatrix} z_2 & z_4 \\ 1 & 1 \end{pmatrix}}{\det\begin{pmatrix} 1 & z_4 \\ 0 & 1 \end{pmatrix} \det\begin{pmatrix} z_2 & z_3 \\ 1 & 1 \end{pmatrix}} = \frac{[u_1, u_3][u_2, u_4]}{[u_1, u_4][u_2, u_3]} = (u_1, u_2, u_3, u_4).$$

This is quite miraculous. The cross ratio is not only invariant under ϕ but also m_ϕ. In addition to this, the definition of the cross ratio in terms of homogeneous coordinates is more unifying in the sense that the case of "infinity" is included in the definition. We utilize this point of view to define a cross ratio in several complex variables.

1.4. Linear Fractional Maps and Associated Matrices in Several Complex Variables. We next say what it means to be a linear fractional map in $\mathbb{C}^N$ for $N > 1$. In particular, realize that one cannot just divide by $cz + d$ as z is now a vector, not a scalar. If one takes the position that the associated matrix of a LFM should act as a linear transformation on the homogeneous coordinates as we observed in $\mathbb{C}$, one arrives at the following definition.

Definition 1.3. We say ϕ is a linear fractional map in $\mathbb{C}^N$ if

$$\phi(z) = \frac{Az + B}{\langle z, C \rangle + D}. \tag{1.5}$$

where A is an $N \times N$ matrix, B and C are column vectors in $\mathbb{C}^N$, $D \in \mathbb{C}$, and $\langle \cdot, \cdot \rangle$ is the standard inner product.

We define the associated matrix m_ϕ of the linear fractional map $\phi(z) = \frac{Az+B}{\langle z,C\rangle+D}$ to be given by

$$m_\phi = \begin{pmatrix} A & B \\ C^* & D \end{pmatrix}.$$

Just as in one variable, we have the following. If $\phi(z) = w$ and the point $u \in \mathbb{C}^{N+1}$ is associated with z, then $v = m_\phi u$ is associated with the point w and vice versa. A routine calculation also shows that $m_{\phi_1 \circ \phi_2} = m_{\phi_1} m_{\phi_2}$ and $m_{\phi^{-1}} = (m_\phi)^{-1}$. That is, if $z \sim u$, then $\phi(z) \sim m_\phi u$.

Example 1.4. Let ϕ be the linear fractional map in two complex variables given by

$$\phi(z) = \phi(z_1, z_2) = \left(\frac{z_1+1}{-z_1+3}, \frac{2z_2}{-z_1+3} \right).$$

Identifying $\langle z, C\rangle$ with $C^* z$, we can write this as

$$\left(\frac{z_1+1}{-z_1+3}, \frac{2z_2}{-z_1+3} \right) = \frac{\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix}}{(-1,0)^T (z_1, z_2) + 3}.$$

with $A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$, $B = (1,0)^T$, $C = (-1,0)^T$, and $D = 3$ where T represents the transpose of a vector.

2. Propetries of LFMs in $\mathbb{C}^N$

LFMs in $\mathbb{C}^N$ for $N > 1$ share many properties with their one variable counterpart. In this section, we briefly review a few of these. See [**4**], [**5**] for detailed proofs. For information on their applications in operator theory, see [**1**].

Theorem 2.1. *In $\mathbb{C}$, LFMs map generalized circles (meaning circles and lines) to generalized circles.*

This is one of the foundational facts about LFMs in one variable. Below is the several variable analogue.

Theorem 2.2. *In $\mathbb{C}^N$ for $N > 1$, LFMs map generalized ellipsoids to generalized ellipsoids.*

Here a generalized ellipsoid is defined as the image of the unit ball $\{z \mid |z| < 1\}$, where $|z|^2 = |z_1|^2 + \cdots |z_n|^2$, under an invertible linear transformation composed with a translation.

Next we decompose LFMs into their elementary parts. A quick verification shows that whe $c \neq 0$, we have

$$\phi(z) = \frac{az+b}{cz+d} = \frac{a}{c} + \frac{bc-ad}{c} \cdot \frac{1}{cz+d} \tag{2.1}$$

from which it follows:

Theorem 2.3. *In $\mathbb{C}$, LFMs are a composition of expansions, rotations, translations, and inversions.*

To find the $N > 1$ analogue, we define a LFM from $\mathbb{C}^N$ to $\mathbb{C}^N$ to be a **reflection** if its associated matrix is a permutation matrix permuting precisely two variables and fixing the rest and to be **multi-linear** if it can be written in the form $(\langle z, \alpha_1\rangle + c_1, \langle z, \alpha_2\rangle + c_2)$ for some vectors α_1, α_2 and constants c_1, c_2.

Notice that expansions, rotations, and translations are simply multi-linear maps in $\mathbb{C}$ and the inversion $\phi(z) = \frac{1}{z}$ has associated matrix $m_\phi = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, the sole permutation matrix of two variables and is hence a reflection.

THEOREM 2.4. *In $\mathbb{C}^N$ for $N > 1$, LFMs are a composition of multi-linear maps and reflections.*

One may also define cross ratio in several variables. For proof of concept, we do so for $N = 2$. As before, brackets $[\cdot]$ will denote the determinant of the columns inside.

DEFINITION 2.5. Given 5 distinct points u_i for $i = 1, ..., 5$ in $\mathbf{CP}^2$, we define the cross ratio as

$$(u_1, u_2, u_3, u_4, u_5) = \frac{[u_1, u_3, u_5][u_2, u_4, u_5]}{[u_1, u_4, u_5][u_2, u_3, u_5]}. \tag{2.2}$$

This particular definition was chosen so that, using appropriate coordinates, it reduces to the classical cross ratio. However, it's not quite the complete generalization. We want to construct a map from $\mathbb{C}^2$ to $\mathbb{C}^2$. We posit the following definition.

DEFINITION 2.6. We define the cross ratio pair in $\mathbf{CP}^2$ as

$$(u_1, u_2, u_3, u_4, u_5)_2 = \left(\frac{[u_1, u_3, u_5][u_2, u_4, u_5]}{[u_1, u_4, u_5][u_2, u_3, u_5]}, \frac{[u_1, u_3, u_4][u_2, u_4, u_5]}{[u_1, u_4, u_5][u_2, u_3, u_4]} \right) \tag{2.3}$$

Now if z_1 is the point associated with u_1, just as in the case of one variable, setting $z_1 = z$, a variable implies this defines a linear fractional map in $\mathbb{C}^2$!

One important note to the astute reader. Recall that for homogeneous coordinates, for a scalar c, the point (u_1, u_2) is associated with the same point as $c(u_1, u_2)$. Hence there are an infinite number of representatives (all lying on the same complex line) for a given $z \in \mathbb{C}$. The subscripts on these cross ratio definitions are crafted precisely so that it is independent of the representative chosen. For every c showing up in the numerator, it also appears in the denominator and they cancel out!

With such a definition, we have a 3-transitive analog in higher dimensions.

THEOREM 2.7. *LFMs in $\mathbb{C}^N$ are $(N+2)$-transitive but they are not $(N+3)$-transitive.*

One caveat to this theorem is as follows. Recall that in $\mathbb{C}$, our three points had to be distinct. In $\mathbb{C}^N$, the vectors in $\mathbf{CP}^N$ associated with our points in $\mathbb{C}^N$ have to be linearly independent. Under this condition, the above theorem holds.

3. Open Problems for the Enthusiastic Reader

LFMs in $\mathbb{C}$ are one of the most well-studied class of maps in complex analysis. While the author has taken time to establish some of the most basic properties of our LFMs in $\mathbb{C}^N$ for $N > 1$, a world of open problems await. Any established properties of LFMs in $\mathbb{C}$ that strike the reader's fancy and have not been noted here are likely open to generalization. Here just a few that have piqued the author's interest.

- A cross ratio has been defined for several variables [**4**]. Similarly, a cross ratio (in one variable) has been defined for quaternions [**2**]. [**3**]. Can one define a cross ratio in several quaternionic variables? Why or why not? If so, what are its properties?
- The cross ratio in one variable is used to define distance in hyperbolic geometry (think M.C. Escher paintings) via the Cayley-Klein metric. Does an analogue exist in higher dimensions?
- In [**5**], it is determined when a LFM in $\mathbb{C}^N$ is a self-map of the unit ball, defined by $\{z \mid |z| < 1\}$. When is a LFM a self-map of the unit polydisk defined by $\{z = (z_1, ..., z_n) \mid |z_j| < 1, \forall 1 \leq j \leq n\}$?

The reader is also encouraged to open her favorite complex variables textbook, look at the section on LFMs along with the corresponding exercises, and explore what properties remain to be generalized!

References

[1] Cowen, Carl C., and Barbara D. MacCluer. "Linear fractional maps of the ball and their composition operators." Acta Sci. Math.(Szeged) 66.1-2 (2000): 351-376.

[2] Gwynne, Ewain, and Matvei Libine. "On a quaternionic analogue of the cross-ratio." Advances in Applied Clifford Algebras 22.4 (2012): 1041-1053.

[3] Libine, Matvei, and Surya Raghavendran. "Geometric properties of conformal transformations on R p, q." Geometriae Dedicata 178.1 (2015): 323-335.

[4] Pilla, Michael R. "A generalized cross ratio". Contemporary Mathematics Volume 799, 2024. Recent Progress in Function Theory and Operator Theory 799 (2024): 41.

[5] Pilla, Michael R. "Linear fractional self-maps of the unit ball." Canadian Mathematical Bulletin 67.2 (2024): 458-468.

Florida Polytechnic University
Current address: Florida Polytechnic University
E-mail address: `mpilla@floridapoly.edu`